

Partial Differential Equations 3 (Numerical)

Lecture 10 - Academic Year 2025/2026

Prof Halim Kusumaatmaja
Email: halim.kusumaatmaja@ed.ac.uk
Semester 2, Edinburgh Engineering

Aims and Objectives

1. Lax-Wendroff TVD Scheme.
2. Lax-Wendroff TVD in two dimensions.

Rewriting Lax-Wendroff Scheme Using Fluxes

From the previous lecture, recall that the Lax-Wendroff scheme can be written as

$$u_i^{n+1} = u_i^n - \frac{c\Delta t}{2\Delta x}(u_{i+1}^n - u_{i-1}^n) + \frac{c^2\Delta t^2}{2\Delta x^2}(u_{i+1}^n - 2u_i^n + u_{i-1}^n). \quad (1)$$

To adapt the Lax-Wendroff scheme into a TVD method, it is useful to rewrite it in terms of fluxes,

$$u_i^{n+1} = u_i^n + \frac{\Delta t}{\Delta x}(F_{i-\frac{1}{2}}^n - F_{i+\frac{1}{2}}^n). \quad (2)$$

$F_{i+\frac{1}{2}}^n$ and $F_{i-\frac{1}{2}}^n$ have the same form. However, $F_{i+\frac{1}{2}}^n$ involves grid points i and $i+1$, while $F_{i-\frac{1}{2}}^n$ involves grid points $i-1$ and i .

Exercise 1: The Lax-Wendroff flux is given by

$$F_{i-\frac{1}{2}}^n = cu_{i-1}^n + \frac{1}{2}c \left(1 - \frac{\Delta t}{\Delta x}c\right) [u_i^n - u_{i-1}^n], \quad (3)$$

$$F_{i+\frac{1}{2}}^n = cu_i^n + \frac{1}{2}c \left(1 - \frac{\Delta t}{\Delta x}c\right) [u_{i+1}^n - u_i^n]. \quad (4)$$

By substituting these into Eq. (2), show that we can reproduce the Lax-Wendroff scheme in Eq. (1).

Lax-Wendroff TVD Scheme

To make the Lax-Wendroff TVD, we can change the flux definition. In general, this can be written as:

$$F_{i-\frac{1}{2}} = FL_{i-\frac{1}{2}} + \phi(r_{i-\frac{1}{2}}) [FH_{i-\frac{1}{2}} - FL_{i-\frac{1}{2}}], \quad (5)$$

$$F_{i+\frac{1}{2}} = FL_{i+\frac{1}{2}} + \phi(r_{i+\frac{1}{2}}) [FH_{i+\frac{1}{2}} - FL_{i+\frac{1}{2}}]. \quad (6)$$

$\phi(r)$ is the limiter function and r is a measure of the smoothness of the solution. The limiter function is designed to be equal to one in regions where the solution is smooth, in which F reduces to FH , corresponding to a high order scheme. In regions where the solution is not smooth (i.e. in regions containing sharp gradients and discontinuities), the limiter function is designed to be equal to zero, in which F reduces to FL , corresponding to a low order scheme. As a measure of the smoothness, r is commonly taken to be the ratio of consecutive gradients

$$r_{i-\frac{1}{2}} = \frac{u_{i-1} - u_{i-2}}{u_i - u_{i-1}}; \quad r_{i+\frac{1}{2}} = \frac{u_i - u_{i-1}}{u_{i+1} - u_i}. \quad (7)$$

A flux limited version of the Lax-Wendroff scheme could thus be obtained by adding a limiter function ϕ to the second terms in Eqs. (8) and (9).

$$F_{i-\frac{1}{2}}^n = cu_{i-1}^n + \phi(r_{i-\frac{1}{2}})\frac{1}{2}c\left(1 - \frac{\Delta t}{\Delta x}c\right)[u_i^n - u_{i-1}^n], \quad (8)$$

$$F_{i+\frac{1}{2}}^n = cu_i^n + \phi(r_{i+\frac{1}{2}})\frac{1}{2}c\left(1 - \frac{\Delta t}{\Delta x}c\right)[u_{i+1}^n - u_i^n]. \quad (9)$$

We can immediately see that when $\phi = 0$ we get the FOU scheme, while when $\phi = 1$ we get back the original Lax-Wendroff scheme.

The question now is how to define the limiter function that ensures we get a TVD scheme. Sweby derived that to be TVD, the limiter function must be in the shaded region in Fig. 1. In the literature there are a large number of possible limiter functions that have been proposed. Note that in all these cases $\phi(< 0) = 0$.

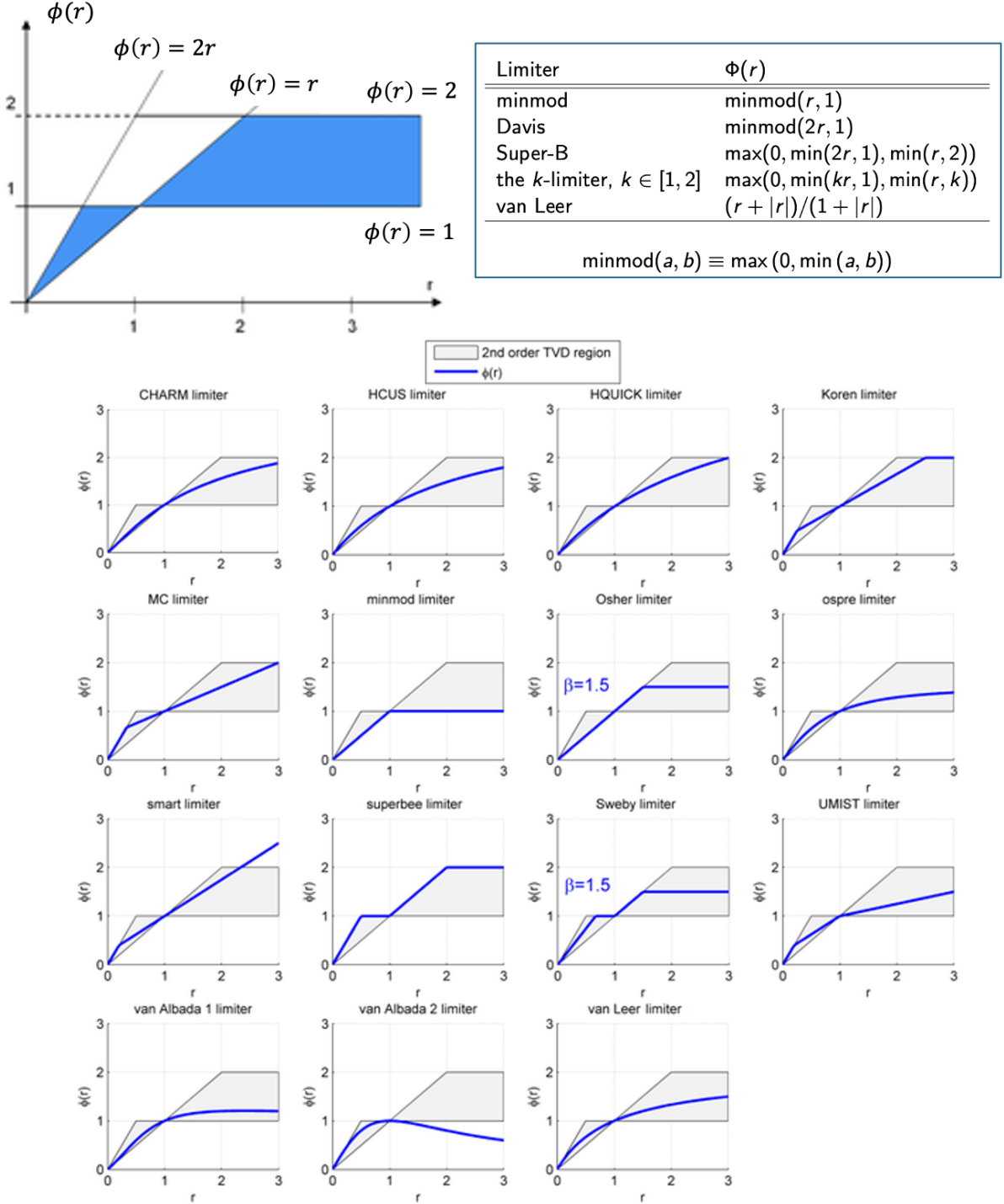


Figure 1: Sweby's admissible limiter region for second-order TVD schemes.

Fig. 2 shows a comparison of the simulation results when we use the original Lax-Wendroff scheme and Lax-Wendroff TVD scheme. In this example, we have studied once again the test problem provided in the previous lecture, where the solution of interest has both a smooth feature and a step function. The Lax-Wendroff TVD scheme is more accurate compared to the FOU scheme (see Lecture 9), and at the same time, it avoids the problem with large spurious oscillation at the discontinuities observed for the original Lax-Wendroff scheme. Nonetheless, we can see that the result is still not perfect. This illustrates the difficulties in solving hyperbolic PDEs numerically when there are discontinuities, which remains an active area of research in the field of numerical PDEs. There are numerous more advanced schemes which are beyond the scope of our course!

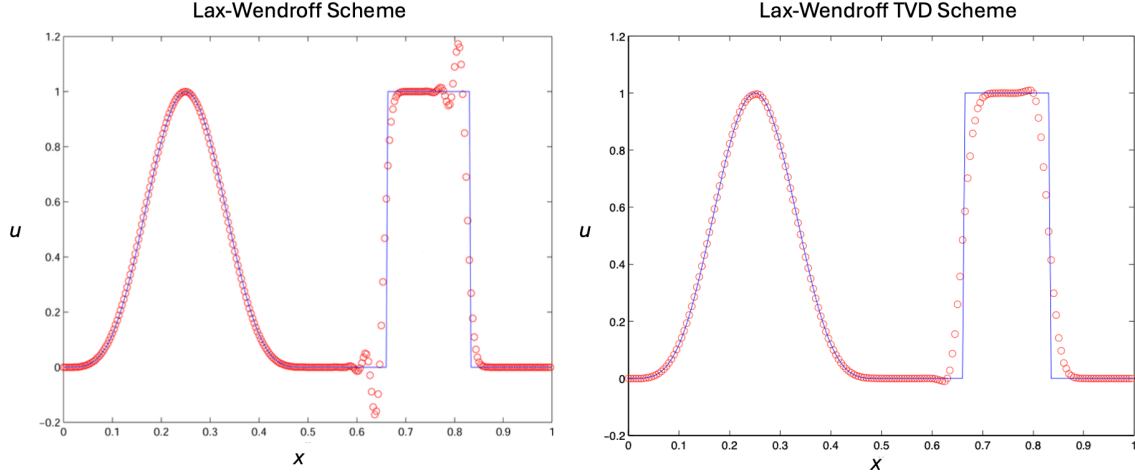


Figure 2: A comparison between the original Lax-Wendroff and Lax-Wendroff TVD schemes.

Beyond One Spatial Dimension

Most engineering problems are not in one dimension, rather in two and three dimensions. Hence, there is a clear need to generalise the one dimensional methods. In particular, it is advantageous if a complex two or three dimensional problem can be reduced to a sequence of one dimensional problems. The method of fractional steps was developed to address this problem, akin to what we have discussed in Lecture 8 for the Crank-Nicolson method. A further benefit of this approach is that the time step allowed for one dimensional schemes is generally larger than that allowed for multi-dimensional problems.

Consider the following 2-D advection equation

$$\frac{\partial u}{\partial t} + c_x \frac{\partial u}{\partial x} + c_y \frac{\partial u}{\partial y} = 0. \quad (10)$$

As in Lecture 9, performing a Taylor expansion in time gives us

$$u(x, y, t + \Delta t) = u(x, y, t) + \Delta t \partial_t u(x, y, t) + \frac{\Delta t^2}{2!} \partial_{tt} u(x, y, t) + O(\Delta t^3). \quad (11)$$

Given the advection equation we consider, $\partial_t u = -c_x \partial_x u - c_y \partial_y u$. Moreover, differentiating the advection equation with respect to t ,

$$\partial_{tt} u = c_x^2 \partial_{xx} u + (c_x c_y \partial_x \partial_y + c_y c_x \partial_y \partial_x) u + c_y^2 \partial_{yy} u.$$

Substituting these relations to Eq. (11), we obtain

$$u(x, y, t + \Delta t) = \hat{L}_{xy} u(x, y, t), \quad (12)$$

$$\hat{L}_{xy}(\Delta t) = \left[1 - \Delta t (c_x \partial_x + c_y \partial_y) + \frac{\Delta t^2}{2} (c_x^2 \partial_{xx} + (c_x c_y \partial_x \partial_y + c_y c_x \partial_y \partial_x) + c_y^2 \partial_{yy}) \right]. \quad (13)$$

The key question now is whether we can write $\hat{L}_{xy}$ in terms of $\hat{L}_x$ and $\hat{L}_y$. Based on what we have done in Lecture 9, we know that

$$\hat{L}_x(\Delta t) = 1 - c_x \Delta t \partial_x + \frac{c_x^2 \Delta t^2}{2} \partial_{xx}. \quad (14)$$

$$\hat{L}_y(\Delta t) = 1 - c_y \Delta t \partial_y + \frac{c_y^2 \Delta t^2}{2} \partial_{yy}. \quad (15)$$

Exercise 2: Show that $\hat{L}_{xy}(\Delta t) \neq \hat{L}_x(\Delta t)\hat{L}_y(\Delta t)$ if $\partial_x \partial_y$ does not commute with $\partial_y \partial_x$. This typically happens if the function u is continuous but its derivative is not continuous.

The results in Exercise 1 means that if we replace $\hat{L}_{xy}(\Delta t)$ with $\hat{L}_x(\Delta t)\hat{L}_y(\Delta t)$, we may only get a first order accurate scheme in time. This is undesirable.

The good news is that the following sequence of operators allows us to replace the 2D operator with a series of 1D operators

$$\hat{L}_{xy}(\Delta t) = \hat{L}_x\left(\frac{\Delta t}{2}\right) \hat{L}_y(\Delta t) \hat{L}_x\left(\frac{\Delta t}{2}\right). \quad (16)$$

In fact, any symmetric sequence may be used, such that over N time steps we can employ the following sequence of operators

$$\hat{L}_x\left(\frac{\Delta t}{2}\right) \hat{L}_y(\Delta t) \hat{L}_x(\Delta t) \cdots \hat{L}_y(\Delta t) \cdots \hat{L}_x(\Delta t) \hat{L}_y(\Delta t) \hat{L}_x\left(\frac{\Delta t}{2}\right). \quad (17)$$

The advantages of this approach are as follows:

- The one dimensional operators are usually easier to implement.
- A time step which is stable for the 1D scheme can be used, this is typically twice that allowed by the 2D scheme (more for 3D).
- The factored sequence is easily extended to include other effects and more complex terms of the equations.

Additional Materials:

- Sweby, P.K. (1984), "High resolution schemes using flux-limiters for hyperbolic conservation laws", SIAM J. Numer. Anal., 21 (5): 995–1011

Version Control:

- version 1.0 on 20 December 2025.